Statistical Analysis Plan (SAP)
External Controls for XB010, an experimental agent for subjects with locally advanced or metastatic solid tumors

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LIST OF ABBREVIATIONS

Abbreviation	Description of Abbreviated Term
SAP	Statistical analysis plan
N-POWER MEDICINE	N-Power Medicine
RCT	Randomized control trial
ECA	External control arm
NSCLC	Non-small cell lung cancer
ORR	Objective response rate
HIPAA	Health Insurance Portability and Accountability Act of 1996
RECIST 1.1	Response Evaluation Criteria in Solid Tumors version 1.1
PS	Propensity score
ATT	Average treatment effect among the treated
ATE	Average treatment effect
IPTW	Inverse-probability of treatment weighted
KM	Kaplan-Meier
DOR	Duration of response
PFS	Progression-free survival
OS	Overall survival
BOR	Best overall response
CR	Complete response
PR	Progression of disease
SD	Stable disease
NE	Not evaluable
ECOG	Eastern Cooperative Oncology Group
AE	Adverse Event
LoT	Line of therapy
SoC	Standard of care
CTCAE	Common Terminology Criteria for Adverse Events
CCI	Charlson Comorbidity Index
FFPE	Formalin-Fixed Paraffin-Embedded
DICOM	Digital Imaging and Communications in Medicine
ANC	Absolute Neutrophil Count
AST	Aspartate Aminotransferase
ALT	Alanine Aminotransferase
ULN	Upper limit of normal
G-CSF	Granulocyte-colony stimulating factor
IO	Immuno-therapy
SACT	Systemic anti-cancer therapy
PC	Physician’s choice
1L	First-line therapy
2L	Second-line therapy

3L	Third-line therapy

1. INTRODUCTION

Randomized controlled trials (RCTs) are designed to assess efficacy and safety of new investigational therapies. However, in some disease settings, conducting RCTs may be infeasible or ethically challenging, which may stem from small patient populations, lack of clinical equipoise, inability to blind treatment arms, or limited treatment options with poor prognosis.

One approach to overcoming these challenges is the use of external control arms (ECAs), which are generated using data external to the clinical trial. ECAs aid in the assessment of investigational therapies by providing a comparator arm for single-arm trials, enabling the evaluation of relative response rates and survival benefits to support decision-making. This statistical analysis plan (SAP) describes the selection of external controls and proposes analyses comparing them to the participants of the XB010-101, Expansion Study, assessing XB010-101as a single agent in participants with metastatic non-small cell lung cancer (NSCLC).

This study will prospectively generate external control patient data that can ensure completeness of key baseline variables and standardization of variables of interest with imaging (modality, coverage, and intervals) like the clinical trial addressing the key weaknesses of retrospective approaches. The study will enroll patients across multiple indications in first, second and third line and beyond metastatic NSCLC where there is a need for high quality external control patients for drug development purposes. Capturing a wide range of variables relevant to the eligibility criteria in clinical trials will enable selecting appropriate external control patients for XB010. The inclusion of Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST1.1) assessments in this study will enable comparisons of objective response rate and progression-free survival.

1.1 STUDY OBJECTIVES

1.1.1. PRIMARY OBJECTIVE

Compare the objective response rate (ORR) as per RECIST1.1 criteria of XB010-101 participants to that of matched external controls.

1.1.2 SECONDARY OBJECTIVES

1. Compare the duration of response as per RECIST1.1 criteria of participants in XB010-101 and matched, external controls.
2. Compare progression-free survival (PFS) as defined per RECIST1.1 criteria of XB010-101 participants and matched, external controls.
3. Compare the overall survival (OS) of XB010-101 participants and matched, external controls.
4. Compare the incidence of select adverse events in XB010-101 participants and matched, external controls.

2. STUDY DESIGN & EXTERNAL CONTROL DATA

2.1 STUDY DESIGN AND STUDY PERIOD

This is a prospective external control cohort study that is observational with respect to treatment selection but standardized with respect to baseline characterization, imaging modality, coverage and scan intervals, systematic RECIST1.1 assessments, select patient data collection and information on treatment discontinuation, in order to address selection bias, measurement bias, and ensure enhanced compatibility of endpoints.

The follow-up will span from the *index date* to the first of study end date on January 31, 2027 or death. The *index date* is defined as the date of first systemic anticancer therapy administration in the line of therapy where the patient is consented to the study.

2.2 SAMPLE SIZE

A sample of 40 external controls and all XB010-101 participants will be available for this analysis.

2.3 EXTERNAL CONTROL DATA SOURCES

Data for the external controls will be sourced from treating physician's eligibility and baseline variable confirmation forms, physician's discontinuation forms, study specific patient questionnaires, electronic medical records, N-Power Medicine's Kaleido Registry, death registry data, pharmacy and medications data, genomic laboratory data, pathology, imaging medical note content, medical imaging source data (in Digital Imaging and Communications in Medicine [DICOM] format) and linkages.

Data for all patients enrolled into this study will be collected by N-Power Medicine's staff and managed in identifiable format within the N-Power Medicine's data platform. Data will span from baseline to data cutoff. For the purposes of study analyses, data will be extracted out of the N-Power Medicine's database and into separate study datasets which will be de-identified in compliance with the Health Insurance Portability and Accountability Act of 1996 (HIPAA).

Structured patient data in HL7, CCDA, or electronic report format from the N-Power Medicine's Network partners and any participating site are electronically ingested into N-Power Medicine's data platform. Unstructured data will be abstracted directly from the EHR or from electronically ingested documents into N-Power Medicine's proprietary electronic data capture application. Patient questionnaire data will also be used. For patients enrolled in the Kaleido Registry, data from the registry may be used. Data for all patients enrolled in this study will be extracted from the N-Power Medicine's data platform and segregated into a study analysis dataset.

2.4 ELIGIBILITY CRITERIA

Inclusion Criteria

Patients will be included if they,

- Have pathologically confirmed locally advanced or *de novo* metastatic or recurrent metastatic NSCLC.
- Must have radiographically measurable disease in the 4 weeks prior to the *index date.*
- At study enrollment have documented evidence of:

- Received first line (1L) treatment with a platinum doublet chemotherapy + immunotherapy (IO); prior to initiating or within 14 days of initiating physician's choice (PC) systematic anti-cancer therapy (SACT) in second line (2L).
- Have *PD-L1* ≥50%; received 1L treatment with single or dual agent IO; prior to initiating or within 14 days of initiating PC SACT in 2L.
- Have *PD-L1*<50%; received 1L treatment with platinum doublet chemotherapy + IO; prior to initiating or within 14 days of initiating PC SACT in 2L.
- Received 1L treatment with osimertinib, osimertinib + chemotherapy or lazertinib + amivantamab; prior to or initiating or within 14 days of initiating PC SACT in 2L.
- Prior to initiating or within 14 days of initiating PC, third or greater line of therapy (3L+).
- Have one (1) formalin-fixed, paraffin-embedded (FFPE) tissue block with ≥25mm^2 tissue surface containing ≥20% tumor (no to minimal necrosis) or up to ten (10) 5µm unstained FFPE tissue slides with ≥20% tumor content and one (1) H&E slide from the same block if feasible. The tumor samples are not sourced from bone.
- Are 18 years of age or older at the time of informed consent and able to sign informed consent.
- Have Eastern Cooperative Oncology Group (ECOG) performance score of 0 or 1 within 4 weeks prior to or 14 days after the *index* date.
- Have adequate organ and marrow function, based upon meeting all the following laboratory criteria within 21 days before the first dose of study treatment:
- Absolute neutrophil count (ANC) ≥ 1500/mm^3 (≥ 1.5 GI/L) without granulocyte, colony-stimulating factor (G-CSF)
- Platelets ≥ 100,000/mm^3 (≥ 100 GI/L)
- Hemoglobin ≥ 9 g/dL (≥ 90 g/L)]
- Aspartate aminotransferase (ALT) and alanine aminotransferase (AST) ≤ 3 × ULN or ≤ 5 with documented liver metastases
- Total bilirubin ≤ 1.5 × ULN (for subjects with Gilbert's disease or unconjugated hyperbilirubinemia ≤ 3 × ULN)
- Calculated creatinine clearance ≥ 45 mL/min (≥ 0.75 mL/sec) using the Cockcroft-Gault equation.

Exclusion Criteria

Patients will be excluded if they,

- Received radiation therapy within the 14 days prior to *index* date
- Had brain metastases that were treated and were NOT stable for at least 4 weeks prior to the *index* date.
- Had severe liver disease as reflected in the Charlson Comorbidity Index.
- Had surgery in the 4 weeks prior to the *index* date.
- Were diagnosed with another malignancy in the 2 years prior to the *index* date, except for superficial skin cancers, or localized, low-grade tumors deemed cured and not treated with systemic therapy.

3. STUDY VARIABLES

3.1 TREATMENTS

Variable Name	Variable Definition
Therapy name (abstracted)	Identifies the generic name of the specific drug listed as treatment for this cancer type in SEER*Rx (https://seer.cancer.gov/seertools/seerrx/).
Therapy class (derived from abstracted)	Antineoplastic therapy class of corresponding therapy name (e.g. immunotherapy, chemotherapy, etc.).
Therapy start date (abstracted)	Start date (year, month, day) of the corresponding therapy.
Therapy end date (abstracted)	End date (year, month, day) of the corresponding therapy.
Surgery pre-metastatic diagnosis	Indicator of receipt of surgery prior to the metastatic diagnosis for patients with recurrent metastatic disease. Will be missing for patients with *de novo* metastatic disease.
Radiation pre-metastatic diagnosis	Indicator of receipt of radiation prior to the metastatic diagnosis for patients with recurrent metastatic disease. Will be missing for patients with *de novo* metastatic disease.
Surgery in the metastatic setting indicator	Indicator of receipt of surgery in the metastatic setting.
Radiation in the metastatic setting indicator	Indicator of receipt of radiation in the metastatic setting.
Neoadjuvant therapy flag (derived from abstracted)	Indicates whether a patient received neoadjuvant treatment on or after NSCLC diagnosis and prior to definitive surgery date. Categorized as Yes; No; Unknown; Not Applicable. Patients who did not undergo surgery will be classified as Not Applicable.
Neoadjuvant therapy regimen (derived from abstracted)	List of neoadjuvant treatment drugs received following NSCLC diagnosis and prior to surgery.
Neoadjuvant therapy start date (derived from abstracted)	Start date (year, month, day) of the first systemic therapy given as part of the neoadjuvant regimen.
Neoadjuvant therapy end date (derived from abstracted)	End date (year, month, day) of neoadjuvant therapy. If a therapy was continued beyond the definitive surgery date, the end date is set to the definitive surgery date.
Adjuvant therapy flag (derived from abstracted)	Indicates whether a patient received adjuvant therapy after definitive surgery and before disease recurrence (local, regional, or distant) or death. Categorized as Yes; No; Unknown; Not Applicable. Patients who did not undergo definitive surgery will be classified as Not Applicable.
Adjuvant therapy regimen (derived from abstracted)	List of adjuvant treatment drugs received.
Adjuvant therapy start date (derived from abstracted)	Start date (year, month, day) of the first systemic therapy given as part of the adjuvant regimen.
Adjuvant therapy end date (derived from abstracted)	End date (year, month, day) of adjuvant therapy. If a treatment continued beyond the end of the adjuvant setting ascertainment window (earliest of recurrence date, progression date, metastatic date), the end date is set to the adjuvant setting ascertainment window end date.
Line name (derived from abstracted)	Line name (1L, 2L, etc.) in the locally advanced or metastatic setting. The lines of therapy are analytically derived based on N-Power Medicine's line of treatment algorithm listed in *Section 5.1.3*.
Line regimen (derived from abstracted)	The regimen name for the corresponding treatment in the locally advanced or metastatic setting.
Line start date (derived from abstracted)	Start date (year, month, day) of the corresponding line of therapy.
Line end date	End date (year, month, day) of the corresponding line of therapy.

(derived from abstracted)	
Index line (Derived)	Indicator of whether the line of therapy denotes the index line of therapy (>= 2L).
Index line of treatment reason for discontinuation	Investigator-stated reason for index line of treatment discontinuation (disease progression, toxicity, etc.)

3.2 CLINICAL OUTCOMES

Radiology images will be acquired for RECIST assessments, but the RECIST assessments will not be reported to the treating physician. A baseline scan is required within 4 weeks prior to the start of the *index* line of therapy, and then every 6 weeks (+/-1) post therapy initiation (i.e., weeks 6, 12, 18, 24, 30, 36). Starting at week 48 the scan interval increases every 12 +2 weeks until disease progression assuming the patient remains on current therapy.

Date of death (structured, abstracted)	The date of death is obtained through abstraction, and through third party data integration. A missing date of death is assumed to imply the patient is alive by the study end date.
Date of last assessment (derived)	The date of the most recent assessment. This date will be used as one of the censoring events for progression-free survival and duration of response.
RECIST1.1 Best overall response (BOR) (derived from BCIR)	The BOR is defined as the best response across all assessment time points. It is the best response a participant has had from the start of the index LoT until progression or the last evaluable assessment in the absence of progression. The BOR is determined according to the following hierarchy: CR ≥ PR ≥ stable disease (SD) ≥ progressive disease (PD) ≥ not evaluable. Both CR and PR will be confirmed based on RECIST1.1, and the minimum interval between 2 assessments will be no less than 4 weeks (28 days). If the minimum time interval is not met, the best response will depend on the subsequent assessments. For participants who are lost to follow-up after an SD assessment not meeting the minimum time criteria, the BOR will be considered NE.
Progression-free survival (months), PFS (derived from BCIR)	Progression-free survival will be defined as the time from the index date to the earliest of either the date of radiographic progression per RECIST1.1 or the date of death due to any cause. Participants alive and progression-free will be censored at the earliest of the date of the last adequate tumor assessment, date of next line of treatment initiation if the patient has no evidence of progression, the date of last assessment if the patient missed two consecutive assessments, study end date. PFS will be calculated as: PFS (months) = (earliest date of progression, death, censoring – index date + 1) / 30.4375
Overall survival (months), OS (derived)	Overall survival will be defined as the time from the index date to the date of death; participants alive by the study end will be censored on the study end date. Overall survival will be calculated as follows: OS (months) = (earliest date of death, censoring – index date + 1) / 30.4375
Duration of response (months), DUR (derived from BCIR)	Duration of response (DOR) will only be assessed among responders, defined as patients that achieved RECIST v1.1 Partial Response (PR) or Complete Response (CR). Duration of response will be defined as the time from the first documentation of an objective response (CR or PR) that is subsequently confirmed at a visit ≥ 28 days later until the earliest of: • Radiographic evidence of disease progression from any cause,

	• Death from any cause Participants who do not have evidence of progression and did not die will be censored at the earliest of: the date of the last tumor assessment, date of last assessment prior to next line of treatment initiation, the date of last assessment if the patient missed two consecutive assessments (patient considered lost to follow-up), study end date. Duration of response will be calculated as follows: Duration of response (months) = (earliest date of progression or death due to any cause or censoring – date of first objective response + 1) / 30.4375
Investigator-reported best overall response (derived from abstracted)	The best overall response to the index line of treatment, as stated by the treating physician at the time of index line of treatment discontinuation.
Investigator-reported date of progression (abstracted)	The first date when the treating physician indicates disease progression during the index line of treatment.

3.3 COVARIATES

Year of birth (derived from structured)	Year of birth.
Age at initial NSCLC diagnosis (derived)	Age at initial cancer diagnosis for patients diagnosed in earlier stages that experienced metastatic recurrence. Will be missing for patients with *de novo* metastatic disease.
Age at index (derived)	Age at the index LoT initiation.
Sex (structured)	Sex at birth.
Race (Patient Questionnaire)	Patient's self-reported race at the time of consent.
Ethnicity (Patient Questionnaire)	Patient's self-reported ethnicity at the time of consent.
Baseline ECOG Performance Status	
Smoking Status (Derived from Patient Questionnaire)	Patient's self-reported smoking status at the time of consent. Categorized as: current, former, never, unknown.
Income (Patient Questionnaire)	Patient's self-reported annual household income at the time of consent.
Histology Group (abstracted)	Tumor histology at the time of cancer diagnosis
Clinical Stage Group at Diagnosis (abstracted)	Clinical stage at initial NSCLC diagnosis defined as the American Joint Committee on Cancer (AJCC) stage group category closest to the NSCLC diagnosis date. Categorized as: "Occult carcinoma"; "0"; "IA"; "IB"; "IIA";"IIB"; "IIIA";"IIIB"; "IV"; "Not Available"
Clinical T Value at Diagnosis (abstracted)	The clinical T value corresponding to the clinical stage group. Categorized as: "TX"; "T0"; "Tis"; 'T1"; "T1a"; "T1b"; "T2"; "T2a"; "T2b"; "T3"; "T4"; "Not Available".
Clinical N Value at Diagnosis (abstracted)	The clinical N value corresponding to the clinical stage group. Categorized as: "NX"; "N0"; "N1"; "N2"; "N3"; "Not Available"
Clinical M Value at Diagnosis (abstracted)	The clinical M value corresponding to the clinical stage group. Categorized as: "M0"; "M1"; "M1a"; "M1b"; "Not Available"
Pathologic Stage Group at Diagnosis (abstracted)	Stage at initial NSCLC diagnosis, defined as the American Joint Committee on Cancer (AJCC) stage group category closest to the

	NSCLC diagnosis date. Categorized as II; IIA; IIB; III; IIIA; IIIB; Unknown.
Pathologic T Value at Diagnosis (abstracted)	T stage value associated with the stage group.
Pathologic N Value at Diagnosis (abstracted)	N stage value associated with the stage group.
Pathologic M Value at Diagnosis (abstracted)	M stage value associated with the stage group.
Tumor grade (abstracted)	The highest tumor grade found on any pathology report related to the NSCLC primary. The grade will be from the primary site only. In NSCLC, tumor grade refers to the degree of differentiation of cancer cells, indicating how abnormal they are. Categorized as: 'Unknown' if not reported. "GX: Grade of Differentiation Cannot Be Assessed"; "G1: Well-Differentiated"; "G2: Moderately-Differentiated"; "G3: Poorly-Differentiated"; "G4: Undifferentiated"
Coronary artery disease (CAD) (derived from abstracted)	Time-updated indicator of CAD. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Cerebrovascular disease (CVD) (derived from abstracted)	Time-updated indicator of CVD. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Peripheral vascular disease (PVD) (derived from abstracted)	Time-updated indicator of PVD. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Cardiac rhythm disorders (CRD) (derived from abstracted)	Time-updated indicator of CRD. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Heart failure (HF) (derived from abstracted)	Time-updated indicator of HF. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Venus thromboembolic disease (VTD) (derived from abstracted)	Time-updated indicator of VTE. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
AIDS/HIV (derived from abstracted)	Time-updated indicator of AIDS/HIV. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Neurologic disease (ND) (derived from abstracted)	Time-updated indicator of ND. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Other malignancy (not related to index cancer) (derived from abstracted)	Time-updated indicator of a non-index malignancy. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Respiratory disease (derived from abstracted)	Time-updated indicator of respiratory disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Diabetes (derived from abstracted)	Time-updated indicator of diabetes. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Severe liver disease (derived from abstracted)	Time-updated indicator of severe liver disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Gastrointestinal disease (derived from abstracted)	Time-updated indicator of gastrointestinal disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.

Renal disease (derived from abstracted)	Time-updated indicator of renal disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Rheumatic disease (derived from abstracted)	Time-updated indicator of rheumatic disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Musculoskeletal disease (derived from abstracted)	Time-updated indicator of musculoskeletal disease. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Autoimmune and immune-mediated (derived from abstracted)	Time-updated indicator of autoimmune and immune-mediated. Baseline ascertained in the 12 months prior to index; variable is updated if the patient first experiences after index.
Charlson Comorbidity Index Score (derived)	Charlson Comorbidity Index (CCI) using Quan 2011 weights (derived from structured EHR). Calculated based on the weighted sum (using Quan 2011 weights) of the presence of all of the following 17 conditions in the 12 months prior to the mNSCLC diagnosis date, calculated based on structured EMR feeds: AIDS/HIV; dementia (e.g. Alzheimer's); cerebrovascular disease; chronic pulmonary disease; congestive heart failure; diabetes with chronic complications; diabetes without chronic complications; hemiplegia or paraplegia; mild liver disease; metastatic solid tumor (other than primary, if appropriate); moderate or severe liver disease; myocardial infarction; peripheral vascular disease; peptic ulcer disease; renal disease; rheumatic disease; any malignancy (not related to primary cancer type). This updated version of the CCI assigns a weight of 0 to five (myocardial infarction; peripheral vascular disease; cerebrovascular disease; peptic ulcer disease; diabetes without chronic complications) of the original 17 conditions, effectively resulting an an index that is based on 12 conditions. Categorized as {numeric value}; Unknown.
Number of Metastatic Sites (abstracted)	Number of metastatic sites at index.
Brain Metastasis Flag (abstracted)	Indicator of brain metastases at index.
Liver Metastasis Flag (abstracted)	Indicator of liver metastases at index.
Bone Metastasis Flag (abstracted)	Indicator of bone metastases at index.
Reported Adverse Event (abstracted)	The verbatim physician stated AE. Also includes comorbidities collected as part of the Charlson Comorbidity Index. Comorbidities that appear after the index date are considered AE.
Body System or Organ Class (abstracted)	AE Body System or Subcategory. Categorized as: "ILD/Pneumonitis"; "QTc prolongation"; "Heart Failure/Cardiomyopathy"; "Rash"; "Diarrhea"; "Paronychia"; "Infusion- Related Reactions (IRRs)"; "Hematologic Toxicities (neutropenia, etc)"; "Peripheral Neuropathy"; "Hypoalbuminemia"; "Electrolyte Abnormalities"; "Stomatitis/Mucositis"; "Musculoskeletal Pain"; "Dyspnea"; "Nausea"; "Fatigue"; "Edema"; "Constipation"; "Vomiting"; "Thromboembolism"; "Hepatic"; "Renal"; "Endocrine"; "Neurological (encephalitis, etc)", "Other {text}".
Category for Adverse Event (abstracted)	Whether the AE being reported is related to surgery, radiation, or systemic anti-cancer therapy (SACT).
Severity/Intensity (abstracted)	Indicator whether or not this medication has been discontinued or held due to AE. If a drug is held for >60 days, then the medication is recorded as discontinued

Standard Toxicity Grade (abstracted)	The grade of the toxicity if reported. Categorized as: Grade 1 (Mild), Grade 2 (Moderate), Grade 3 (Severe), Grade 4 (Life-threatening), and Grade 5 (Death).
Action Taken with Study Treatment (abstracted)	Indicator of whether or not the medication has been discontinued or held due to AE. If a drug is held for >60 days then the medication is recorded as discontinued, "Treatment Discontinued", "Treatment Held"
AE Start Date (Numeric) (abstracted)	The date at which the physician-stated AE or a new comorbidity was first recorded in a patient's EHR.
AE End Date (Numeric) (abstracted)	The date the verbatim physician stated AE was stated as resolved or ended.
AE Duration (Days) (derived)	AE End Date – AE Start Date + 1;
Serious AE requiring Hospitalization Flag (abstracted)	Indicator of whether the patient was hospitalized for this specific AE. If hospitalization is not mentioned in the EHR, set to 'Unknown'.
Hospitalization admission date (abstracted)	The date on which the patient was admitted to the hospital for the treatment of the AE.
Hospitalization discharge date (abstracted)	The date on which the patient was discharged from the hospital for the treatment of the AE.
Hospitalization Duration (Days) (derived)	Hospitalization discharge date – hospitalization admission date +1.
LOH Summary Statement (abstracted)	Loss of heterozygosity (LOH) as stated in the biomarker report. Categorized as: Low, High, Indeterminate, Not Stated.
LOH Score (abstracted)	LOH score as stated in the biomarker report; the units are percent (%).
Tumor Mutational Burden (TMB) (abstracted)	TMB as stated in the biomarker report. Expressed in mutations/megabase (mut/Mb).
Microsatellite Instability Status (MSI) (abstracted)	MSI status as stated in biomarker report. Categorized as: MSI-HIGH, MSS (microsatellite stable), NOT TESTED.
PD-L1 Summary (abstracted)	PD-L1 status as stated in the biomarker report. Categorized as: High Expression, Low Expression, Positive (Expressed, NOS), Negative (No Expression), Indeterminate, Not Stated.
PD-L1 Score (abstracted)	The stated expression value in the report; the unit is percent (%).
PD-L1 Score Type (abstracted)	Indicates the scoring system for PD-L1 results. Categorized as: Tumor Proportion Score (TPS), Combined Positive Score (CPS), Immune Cell Positivity (IC), and Tumor Cell Positivity (TC). Will be set to missing if patients did not receive PD-L1 testing.
Overall MMR expression (abstracted)	Overall MMR expression as stated in biomarker report. Categorized as: Proficient, Deficient, Not Stated, Other.
MMR MLH1 Expression Status (abstracted)	MMR MLH1 expression as stated in biomarker report. Categorized as: Positive/Intact, Negative/Absent.
MMR MSH2 Expression Status (abstracted)	MMR MSH2 expression as stated in biomarker report. Categorized as: Positive/Intact, Negative/Absent
MMR MSH6 Expression Status (abstracted)	MMR MSH6 expression as stated in biomarker report. Categorized as: Positive/Intact, Negative/Absent
MMR PMS2 Expression Status (abstracted)	MMR PMS2 expression as stated in biomarker report. Categorized as: Positive/Intact, Negative/Absent
ERBB2/HER2 Mutation Status (abstracted)	ERBB2/HER2 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, Variant of unknown significance (VUS), Not Tested.
HER2 IHC (abstracted)	HER2 IHC score as stated in medical record. Categorized as: 0, 1+, 2+, 3+, Quantity not sufficient (QNS), Indeterminate.

HER2 ISH/FISH Amplification (abstracted)	HER2 ISH/FISH status as stated in the medical record. Categorized as: Amplified, Not Amplified, Equivocal, QNS, Indeterminate.
ALK Rearrangement Status (abstracted)	ALK rearrangement status as stated in the biomarker report. Categorized as: Positive, Negative, Not Tested.
EGFR Mutation Status (abstracted)	EGFR rearrangement status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
ERBB2/HER2 Mutation Status (abstracted)	ERBB2/HER2 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
KRAS Mutation Status (abstracted)	KRAS mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
MET Mutation Status (abstracted)	MET mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
NTRK 1 Mutation Status (abstracted)	NTRK 1 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
NTRK 2 Mutation Status (abstracted)	NTRK 2 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
NTRK 3 Mutation Status (abstracted)	NTRK 3 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
RB1 Mutation Status (abstracted)	RB1 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
RET Mutation Status (abstracted)	RET mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
ROS1 Rearrangement Status (abstracted)	ROS1 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
TP53 Mutation Status (abstracted)	TP53 mutation status as stated in the biomarker report. Categorized as: Altered, Not Altered, VUS, Not Tested.
Actionability (derived)	Indicator of the presence of an actionable mutation in KRAS, EGFR, ALK, or ROS1 oncogenes.

4. ECA CONSTRUCTION

The ECA construction process will be conducted on blind-to-outcome data. N-Power Medicine will provide SDTM (Study Data Tabulation Model) datasets containing only demographic and baseline characteristics for patients that meet the inclusion and exclusion criteria. Exelixis will merge these data with XB010-101expansion cohort data.

4.1 PROPENSITY SCORE ESTIMATION

Propensity score (PS) methods are commonly used in observational research to reduce bias from confounding variables that are unbalanced between groups of interest. In RCTs, treatment groups are expected to be balanced with respect to both observed and unobserved baseline covariates due to randomization. PS methods will be used to balance baseline characteristics between ECAs and XB010-101 patients. If balance is achieved for all confounders, differences in the outcomes between the XB010-101 patients and ECAs patients can then reliably be examined.

A PS represents the probability of treatment assignment, conditional on observed pre-treatment characteristics. Balancing algorithms, such as matching or weighting patients based on PS or functions thereof, can be used to construct external controls where treatment assignment is independent of observed baseline covariates. This enables unbiased estimation of treatment effects, assuming no unmeasured confounding. The PS is defined as the probability of a patient being assigned to the XB010-101trial, conditional on their baseline characteristics (i.e., potential confounders). That is,

$$e(x) = P(T = 1|\text{X=x})|)$$

where $T = 1$ if a patient belongs to the XB010-101trial group or $T = 0$ if the patient belongs to external controls; and X is a vector representing the covariates to be included in the propensity score model. A propensity score model will be estimated using logistic regression. The predicted probabilities from this model are the propensity scores. The model will include stage at initial NSCLC diagnosis (*de novo* metastatic vs. recurrent), number of metastatic sites, and lines of treatment received prior to the index line of treatment. If data allows, the model may additionally include sex, age, histology and other covariates.

Table PO1.0. Odds ratios for the propensity score model.

The absolute standardized difference in covariate means will be computed and compared. In general, an absolute standardized difference of less than 0.25 is considered acceptable (Stuart and Rubin 2007), while a threshold below 0.10 indicates a negligible difference in the mean or prevalence of a covariate between groups (Normand et al. 2001). A histogram will describe the spread of estimated weights. If multiple propensity score models are evaluated, the model that most effectively meets this balance threshold will be selected for subsequent analyses.

5. ANALYSES

Descriptive statistics will be used to characterize the study groups. For categorical variables, statistics will include the frequency (number of patients (N)) and percentage (%) of total patients observed in each category. For continuous variables, variables will be presented as means and standard deviation (SD), median, 25th and 75th percentiles and ranges (minimum and maximum). The calculation of percentages for each covariate will always include the missing category in the case of missing values. Continuous variables may be categorized into intervals, with the distribution of patients (N, %) provided.

The Kaplan-Meier Product Limit estimator will be used to assess the distribution of time-to-event real world clinical outcomes and estimate the probabilities of remaining progression-free, and overall survival, as well as the median and estimates at each month of follow-up for each time-to-event outcome. Kaplan-Meier analysis reports will include number of patients with events, censored patients, the number of patients at risk, as well as median, 25th percentile and 75th percentile times to event (in months), 95% CI of the quartiles and ranges (minimum and maximum). The standard errors of the KM estimates will be calculated using Greenwood's formula (the conventional variance estimator for the KM survival function for right-censored data). Confidence intervals will be constructed using the log–log (complementary log–log) transformation to improve interval coverage and ensure that confidence limits remain within the 0-1 interval, which is important for an external control sample size of 40. Where appropriate, the reverse Kaplan-Meier estimator, which reverses the censoring and event indicators, will be used to describe the distribution of potential follow-up for time-to-event outcomes such as progression-free and overall survival.

De-Identification. To comply with the U.S. Health Insurance Portability and Accountability Act of 1996 (HIPAA), N-Power Medicine transforms and shares data in a manner that meets the de-identification standard under the HIPAA Privacy Rule, as overseen by the Office for Civil Rights, in the U.S. Department of Health and Human Services (U.S. HHS). N-Power Medicine's de-identification policies are supported and approved by Expert Determination. De-identified data is no longer subject to HIPAA, and as such it may be processed and/or commoditized without oversight by HHS.

In order to sufficiently de-identify data such that a patient cannot be re-identified with available public records and other sources, direct identifiers (e.g. name, medical record number) will be excluded from the data transferred where possible and transformed to mask the original values in other cases. All date

variables have been de-identified (shifted) according to Expert Determination as described in the next paragraph. For patients that are alive, years of birth prior to (current year - 89) are set to (current year - 89) so that no patient appears older than 89 since patients 90 or older are more easily identifiable. For deceased patients that died at 89 years of age or later, the year of birth is set to their (year of death - 89).

In order to de-identify dates that are PHI (e.g., date of death, birth), while maintaining the relative relationship between all dates as required for the unbiased estimation of time-to-event endpoints such as overall survival, all dates pertaining to a single patient will be shifted by the same amount. Specifically, each patient is assigned a shift, defined as a random number of days drawn from a Uniform distribution with a minimum of -14, and maximum of 14. All complete original dates (i.e., day, month, and year were known) pertaining to that patient will be then shifted by their individual shift. For example, for a patient that was diagnosed with NSCLC on 1/15/2020, died on 1/15/2022, and whose individual shift was -14 days, the dates of diagnosis and death will appear as 1/1/2020 and 1/1/2022, respectively; their overall survival however, will be 732 days in both cases. While no dates are shifted by more than 2 weeks in either direction, it should be noted that in some rare cases, the shifted dates fall into the following or previous month or year from the original, unaltered date. This caveat should be kept in mind when evaluating temporal trends. For cases when only the month and year of the original date is known, the day will be set to 15, and the shift will not be applied as this would result in an unknown shift of the true date value while exposing the patient-specific shift (e.g., if the shifted day value is 17, the date shift is +2 days). When both the day and month are unknown, the year will be left unaltered.

5.1 EXPLORATORY DATA ANALYSES

The set of table shells and corresponding figure listing that accompany the analyses described in the sections that follow, are provided as a separate document ("{*Table Shells} N-POWER MEDICINE - Exelixis XB0100 ECA SAP.xls*").

5.1.1. Demographic and clinical characteristics.

To assess whether the propensity-score matching has improved comparability between the XB010-101 trial patients and matched controls, we will examine whether the distribution of measured baseline covariates is similar between the XB010-101 trial patients and matched external controls. The set of baseline covariates will include those listed in *Section 3.3*.

Table PO1.1. Baseline characteristics of key demographic and clinical covariates among: (1) XB010-101 participants, (2) the full set of external controls, (3) the propensity score-matched external controls.

5.1.2. Genomic Profile Characterization.

Distributions of test results for the relevant standard of care biomarkers listed in *Section 3.3*., will be evaluated in the (1) XB010-101 trial patients, (2) full set of external controls, and (3) matched external controls. We will calculate the number and percentage of patients with an actionable alteration (or positivity) and the distribution of platform technology used. Where relevant, the distribution of reported scores will be described along with positivity status.

Table PO1.2. Characterization of the genomic profile of: (1) XB010-101 participants, (2) the full set of external controls, (3) the propensity score-matched external controls.

5.1.3. Characterization of Treatment.

Distributions (frequency (number of patients (N)) and percentage (%)) of cancer-related treatments, including surgery, radiation, neoadjuvant and adjuvant treatment for distant recurrent disease, as well as each line of therapy received in the metastatic setting up to and including the index line of treatment will be reported for the (1) XB010-101 participants, (2) the full set of external controls, (3) the set of propensity score matched controls. For groups (2) and (3) results will be further stratified by whether the patients had a relevant actionable mutation.

Table PO1.3. Characterization of the treatment history of: (1) XB010-101 participants, (2) the full set of external controls, (3) the propensity score-matched external controls.

The lines of therapy will be analytically assigned using N-Power Medicine's line of treatment (LoT) algorithm. The algorithm is applied to all antineoplastic drugs that occur either across the entire care journey for a patient's cancer or in a specific setting (e.g., metastatic) within that care journey. The algorithm allows customization of the following three parameters: (1) All drugs initiated within a "window", or number of days from the date of the first drug initiation, define the first line regimen; the value for this parameter is 45 days for NSCLC. (2) Any drugs initiated after a "treatment gap" - the number of days following the discontinuation of all drugs in the previous LoT- constitute a new LoT; the default is 120 days. (3) An indicator of whether "switching" within a LoT is allowed for bioequivalent treatments; this parameter is set to 'no' for NSCLC. Note that only treatments received prior to censoring events should be considered. Specifically, the NSCLC LoT algorithm is as follows:

The first/frontline therapy includes all tumor-specific antineoplastics that start within **45** (window) days of the initiation of the first drug. The first-line:

- *Starts* on the day of the first valid administration (IV only) or first valid order (IV/oral) of antineoplastic agent in the line of therapy.
- *Ends* on the last valid administration of any antineoplastic agent in the line of therapy (LoT) or one of the following end criteria is met.
    - Addition of
        - Switching = 'no': Any new antineoplastic after the first **45** days (window) of a line will trigger the start of a new LoT.
        - Switching = 'yes': Any new antineoplastic after the first **45** days (window) of a line will trigger the start of a new LoT, unless the new antineoplastic is an allowed switch therapy (e.g., bioequivalent agent) for one of the drugs in the line. The allowed switch therapy is added to the line and it continues until another end criterion is met.
    - Treatment gap of >**120** days of all agents in line
        - Switching = 'yes': switch therapies are treated as continuation of therapy in the case of the switch therapy algorithm (i.e., if an allowed switch therapy starts within **120** days (treatment gap) of a therapy ending, then that continues the LoT
    - Death

This algorithm is applied iteratively to all drugs subsequent to the first/frontline to define subsequent LoTs.

5.2 PRIMARY OBJECTIVE: Comparing the ORR in XB010-101 and controls.

5.2.1 Defining the target estimand.

ORR will be defined as the proportion of participants with measurable disease at baseline having a

best overall response (BOR) of confirmed complete response (CR) or confirmed partial response (PR) per RECIST v1.1 by the investigator. The BOR is defined as the best response across all time points. It is the best response a participant has had from the start of the study treatment until progression or the last evaluable assessment in the absence of progression. The BOR will be determined according to the following hierarchy: CR ≥ PR ≥ stable disease (SD) ≥ progressive disease (PD) ≥ not evaluable. Both CR and PR should be confirmed based on RECIST v1.1 and the minimum interval between 2 assessments should be no less than 4 weeks (28 days). If the minimum time interval is not met, the best response will depend on the subsequent assessments. For participants who are lost to follow-up after an SD assessment not meeting the minimum time criteria, the BOR will be considered NE.

The target estimand for this study is the average treatment effect in the treated (ATT), that is, the difference in ORR between XB010-101 participants and the external controls. We denote the quantity of interest as τ where τ > 0 indicates superiority of treatment with XB010, τ < 0 indicates inferiority (or harmfulness) of treatment with XB010-101, and τ = 0 indicates no differences in the effects of treatment with XB010-101and treatment with the standard of care (SoC) received by controls. Formally, we define the ATT as:

$$\tau = E[Y(1) - Y(0)|T = 1]$$

$Y(1)$ is the outcome under treatment with XB010-101$Y(0)$ is the counterfactual outcome under the SOC, and T indicates if a patient belongs to the XB010-101trial group or to external X , and denotes a set of measured covariates. Our estimate of the ATT (and other functions) will be $\hat{\tau}$as . To estimate the ATT in an externally controlled trial, we require the following assumptions:

- Consistency: $Y(t) = Y\ for\ T \in [0,1]$.
- Positivity/Common Support: $0 < P(T = 1|X = x) < 1$ for all x such that $P(X = x) > 0$.
- Conditional Independence/Exchangeability: $Y(t) \perp T|X$.

These assumptions are all typical in causal identification of a treatment effect in randomly controlled trials (RCTs) and met by design of the RCT. Positivity is met by fixing the randomization ratio, and conditional independence is met by randomization. However, in an externally controlled trial, these assumptions are no longer met by design. The latter two assumptions now depend on how similar the external control population is to the single arm population. Violations can lead to inflated type one error and bias in estimation.

5.2.2 Crude Analyses.

The unadjusted ATT is calculated as the difference in mean ORR between XB010-101 participants and external controls, as follows:

$$\widehat{\tau_{crude}} = \frac{1}{n_{XB010}} \sum_{i:T_i=1} Y_i - \frac{1}{n_{EC}} \sum_{i:T_i=0} Y_i$$

Where is the number of XB010-101 trial participants and $n_{EC}$is the number of external controls.

5.2.3 Logistic regression analyses.

The odds ratio for ORR in XB010-101compared to the external controls will be $\exp(\widehat{\beta_1})$where $\widehat{\beta_1}$is

estimated using the following logistic regression,

$$\log i\,t(P(Y_i = 1|T_i,\, X_i|) = \beta_0 + \beta_1 T_i + \beta^{\mathrm{T}} X_i) \quad (1).$$

5.2.4 Inverse probability of treatment weighting (IPTW).

As previously noted, the propensity scores $e(X) = P(T = 1|X)$ are computed by regressing trial participation on the baseline covariates. We then obtain ATT (average treatment effect on the treated) weights using weights $w = 1$ for the XB010-101 participants and

$$w_i = \frac{e(X)}{1 - e(X)}$$

weights for the external control arm participants, where the propensity scores $e(X_i)$ denotes the propensity of being in the external control group for individuals with the set of covariates X_i. . To calculate the propensity score, we will first fit the logistic regression model,

$$\log i\,t\,P(T_i = 1|\, X_i|) = X_i^{\mathrm{T}} \alpha,$$

To estimate coefficients $\hat{\alpha}$. . The propensity scores $\widehat{e_i}$ are then computed as,

$$\widehat{e_i} = \frac{1}{1+\exp(-X_i^T \hat{\alpha})}.$$

If needed, the weights may be truncated at the 99th percentile. Alternatively, no truncation or truncation at the 95th and 90th percentiles may be considered to avoid instability from extreme weights. The ATT is then calculated as the weighted mean ORR difference between XB010-101 participants and external controls, as follows:

$$\widehat{\tau_{iptw}} = \left(\frac{1}{n_{XB010}} \sum_{i:T_i=1} Y_i\right) - \left(\frac{\sum_{i:T_i=0} w_i Y_i}{\sum_{i:T_i=0} w_i}\right)$$

where n_{XB010} is the number of XB010-101trial participants, w_i are the truncated ATT weights, and T_i indicates XB010-101 trial participation, $T_i = 1$ for XB010, $T_i = 0$ for external control.

Finally, we report the ORRs in the two groups, weighted and unweighted differences, and the p-value from a Wald-style weighted two-sample proportion test (Hernan et al., 2001), for the following hypotheses where τ is the true ATT, estimated by the weighted difference in ORRs:

$$H_0\colon \tau \leq 0$$
$$H_a\colon \tau > 0.$$

Additionally, the inverse-probability weighted odds ratio for ORR in XB010-101 compared to the external controls will be estimated using logistic regression (1) and weights w_i.

5.2.5 Propensity score matching.

An alternative to using weighted methods such as IPTW is to match external controls to XB010-101 participants through greedy nearest neighbor matching (Austin 2013) on the propensity score with no replacement and calipers of size 0.2 of the standard deviation of the propensity scores to prevent 'bad' matches (Rosenbaum and Rubin, 1985). A fixed 1-to-1 matching ratio will be used, which aligns with the commonly used 1:1 randomization ratio in clinical trials. Matching will be conducted without replacement; that is, the matched external control patients will be removed from consideration for further matching and the next XB010-101 participant will be selected. Matching will proceed in random order.

XB010 participants without a match within the caliper will be excluded from the matched analysis. Results will be reported for both the full and matched populations. This results in paired data where each match has one XB010-101 participant and one external control. We can thus treat each pair as representing the counterfactual outcome under XB010-101 and SOC. Given the matched data, we can estimate the ATT as the mean difference of the XB010-101 participant and external control ORRs:

$$\widehat{\tau_{matching}} = \frac{1}{m}\sum_{i=1}^{m}\left(\widehat{Y}_i(1) - \widehat{Y}_i(0)\right)$$

where m is the number of matches, $Y_i(1)$ is the outcome for the XB010-101 participant in match *i*, and $Y_i(0)$ is the outcome for the external control in match *i*. For testing, we are again interested in

$$H_0: \tau \leq 0$$
$$H_a: \tau > 0.$$

The primary method for inference will be McNemar's mid-p-value. Conditional logistic regression, conditional on the matched paired can be used as follows,

$$\log i\, t\, P(Y_i(j) = 1) = \alpha_i + \beta X_{ij}$$

Where α_i is the pair-specific effect and $\exp(\beta)$ is the parameter of interest.

Table PO1.4. Objective response rate among (1) XB010-101 participants, (2) the full set of external controls, (3) the propensity score-matched external controls, and respective treatment effects.

If sample sizes permit, the ORR will be additionally evaluated in subgroups of XB010-101participants and external controls defined by the index line number (2L vs. ≥3L), biomarker actionability, or other clinically meaningful factors.

5.3 SECONDARY OBJECTIVES 1, 2 and 3: Comparing the DOR, PFS, and OS of XB010-101 and controls.

Duration of response, overall survival, and progression-free survival are defined in *Section 3.2.* Variations of the Kaplan-Meier estimator will be evaluated to describe the distributions of duration of response, progression-free and overall survival for XB010-101 participants and all available controls. We change the notation in the sections that follow to allow for random variable T to indicate the time to failure. For the following section we define,

- T_i , the failure time (time to progression, or death) for patient i
- δ_i , the censoring indicator (1 = event, 0 = censored)
- A_i = 1 indicates whether a patient belongs to the XB010-101trial, and 0 to external controls.
- $e(X_i) = P(A_i = 1|X_i|)$ is the propensity score, i.e., the probability of group membership given covariates.
- $w_i = \frac{e(X)}{1 - e(X)}$ is the inverse-probability weight.

5.3.1 Unadjusted Kaplan-Meier.

The standard Kaplan-Meier estimator will be used to describe the three time-to-event outcomes (PFS, OS, and DOR among responders) of XB010-101 participants and the external controls. At each time point t ,

the typical Kaplan Meier estimator of the survival curve will be evaluated as follows,

$$\hat{S}(t) = \prod_{u \le t} \left(1 - \frac{\sum_{i:\, T_i = u, \delta_i = 1} 1}{\sum_{i:\, T_i \ge u} 1} \right)$$

Where $\sum_{i:T_i=u,\delta_i=1}^{1}$ is the number of events at time u and $\sum_{i:T_i \ge u}^{1}$ is the number of individuals at risk prior to time u. Standard errors for the Kaplan–Meier estimates will be calculated using Greenwood's formula (Greenwood, 1926). Confidence intervals for the survival function will be constructed using the log–log transformation to ensure appropriate coverage and bounds within the unit interval. Confidence intervals for the median duration of response will be calculated using the Brookmeyer–Crowley method (Brookmeyer and Crowley, 1982), based on inversion of the Kaplan–Meier survival function. Specifically, the lower and upper confidence limits for the median correspond to the earliest time points at which the upper and lower confidence bounds, respectively, for the survival function fall below 0.5. The log-rank test (Mantel, 1966) will be used to compare the survival estimates of XB010-101 participants and external controls. The number of those censored, events, and at risk will be reported for each month of follow-up. All statistical tests will be two-sided with a nominal significance level of α = 0.05, unless otherwise specified.

5.3.2 Inverse probability of treatment weighting (IPTW).

The inverse-probability weighted Kaplan-Meier estimator of the survival function $\widehat{S_{IPTW}}(t)$ will be estimated as follows,

$$\hat{S}_{IPW}(t) = \prod_{u \le t} \left(1 - \frac{\sum_{i:\, T_i = u, \delta_i = 1} w_i}{\sum_{i:\, T_i \ge u} w_i} \right)$$

Where $\sum_{i:T_i=u,\delta_i=1} w_i$ is the weighted number of events, and $\sum_{i:T_i \ge u} w_i$ is the weighted number of individuals at risk prior to time u. Variance estimation will be based on a Greenwood-type estimator applied to weighted risk sets (Austin, 2016). As a sensitivity analysis, robust variance estimation and/or nonparametric bootstrap methods will be used to account for uncertainty in the estimated weights as follows: (1) subjects will be resampled with replacement, (2) propensity scores, weights, and Kaplan-Meier curves will be re-estimated, (3) the variability across bootstrap samples will be calculated. The nonparametric bootstrap fully accounts for propensity score estimation and weight variability and is robust in small samples (Efron and Tibshirani, 1993). Confidence intervals for the survival function will be constructed using a log–log transformation. An adjusted log-rank test can be used to compare the two curves (Xie & Liu 2005).

5.3.3 Propensity score matching.

The standard Kaplan-Meier estimator will be used to evaluate the survival function of the three time-to-event outcomes of the XB010-101 participants and the matched external controls. The distribution of duration of response will be estimated among the subset of matched pairs in which both the XB010-101 participants and the external controls achieved RECIST1.1 partial or complete response. Note tha the propensity score model will be fit using the full analysis population (i.e., all eligible patients in the trial and external control cohorts), irrespective of response status. Propensity score estimation will not be

restricted to responders, as response is a post-baseline outcome and conditioning on response may introduce selection (collider) bias (Rubin and Rosenbaum, 1983).

To account for the correlation within matched sets, variance estimation will be performed using a stratified Greenwood-type estimator (Collett, 2015), which treats each matched set as a stratum. Specifically, the variance contribution at each event time is computed within strata and then combined across strata to obtain the overall standard error. Confidence intervals for the survival function will be constructed using the log–log transformation applied to the stratified Kaplan–Meier estimates, ensuring bounds remain within the unit interval. Sensitivity analyses may use nonparametric bootstrap resampling at the matched-pair level to account for both the matched design and variability in the estimated propensity scores. Comparisons between groups will be performed using stratified log-rank tests (Harrington and Fleming, 1982), where the matched pair is treated as the stratification factor.

The estimates and Kaplan-Meier curves for the groups will be reported in the following table shells,

Table SO1.1. Duration of response (DUR) estimates for XB010-101 participants and external controls who responded to treatment, by estimation approach.

Figure SO1.1. Duration of response estimates for XB010-101 participants and external controls who responded to treatment, by estimation approach.

Table SO2.1. Progression-free survival (PFS) estimates for XB010-101 participants and external controls, by estimation approach.

Figure SO2.1. Progression-free survival (PFS) estimates for XB010-101 participants and external controls, by estimation approach.

Table SO3.1. Overall survival (OS) estimates for XB010-101 participants and external controls, by estimation approach.

Figure SO2.1. Overall survival estimates (OS) for XB010-101 participants and external controls, by estimation approach.

5.4 SECONDARY OBJECTIVE 4: COMPARE THE INCIDENCE OF SELECT ADVERSE EVENTS IN XB010-101 PARTICIPANTS AND MATCHED EXTERNAL CONTROLS.

The incidence of select adverse events listed in *Section 3.3*., will be evaluated among the (1) XB010-101 participants, (2) full set of external controls, and (3) matched external controls. We will calculate the number and percentage of patients who experienced one of the AEs of interest during follow-up, the median and interquartile range of the AE duration in days, the number and percentage of patients for whom the AE was immunotherapy related, the number and percentage of patients that were hospitalized for the AE, the median and interquartile range of the duration of hospitalization, and the number and percentage of patients for whom the AE resolved.

Table SO4.1. Characterization of adverse event incidence and management among: (1) XB010-101 participants, (2) the full set of external controls, (3) the propensity score-matched external controls.

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